

Final_Paper

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Pearson's product-moment correlation

```
data: total_count and mean_level_m
t = -1.4072, df = 8, p-value = 0.197
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8395967  0.2559675
sample estimates:
      cor
-0.4454491
```

Pearson's product-moment correlation

```
data: total_count and mean_level_m
t = 3.5854, df = 8, p-value = 0.007132
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
```

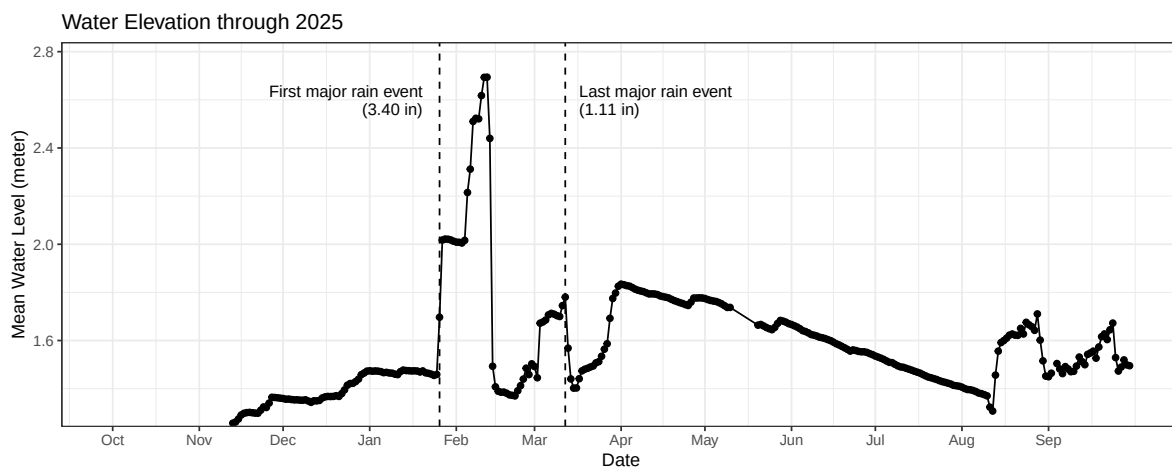
```
0.3074866 0.9467398
sample estimates:
      cor
0.7851136
```

Introduction

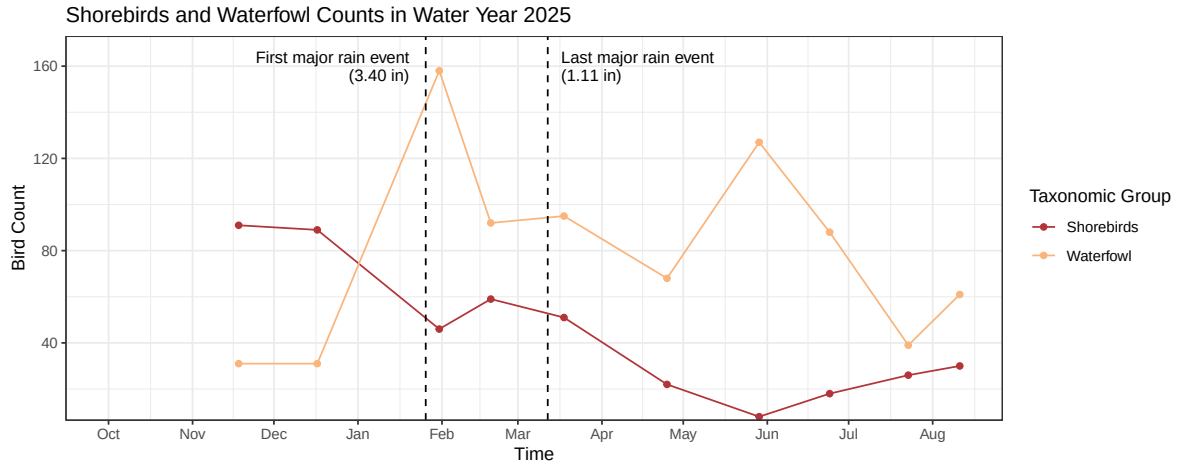
Methods

Results

First and foremost, we explored water elevation and taxonomic abundance as separated variables over time. The first plot shows water elevation through time in the 2025 water year, along with major rain events that significantly spiked water elevation. A limitation for this data is that this water elevation is accurate to local Venoco bridge water elevation, not necessarily the whole wetland.



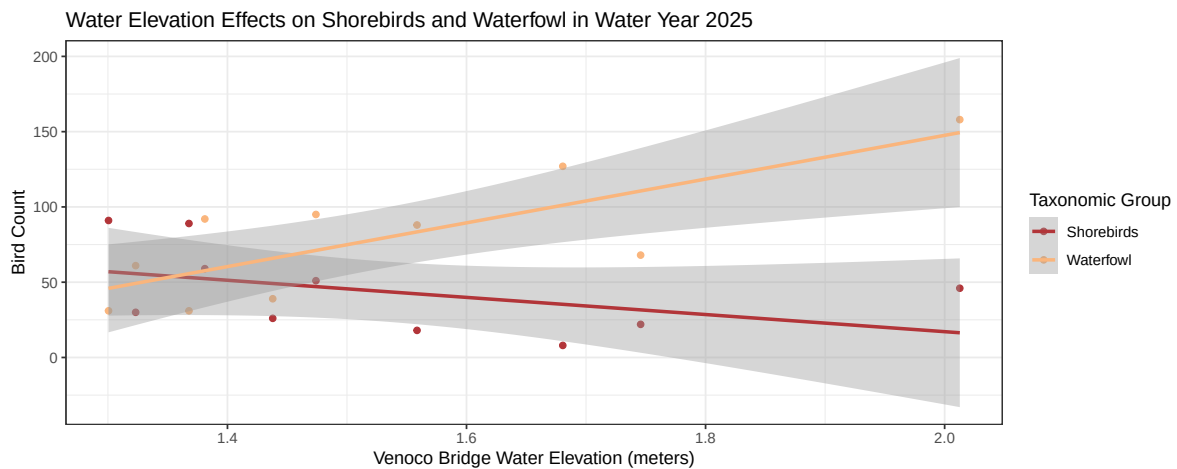
Here you can observe that the highest elevation reached around 2.6 meters with highest elevations concentrating in February 2025. After April, the water levels exhibit a steady decline until August. We then examined how bird counts looked over time within the two taxonomic groups Waterfowl and Shorebirds.



Around the February, Waterfowl showed the highest abundance, adapting well to the high water elevation while its counterpart, the Shorebirds, showed a heavy decline in abundance from high water elevation. Shorebirds continued to decline in numbers, showing a record low right before June before rising back up through to August as water elevation continued to decline.

After reaching their record highest abundance in February, Waterfowl began to decline in March, but suddenly shot back up in abundance in June when Shorebird abundance was almost zero. The plot highlights sharp deviation between the two taxonomic groups, possibly revealing that water elevation affects them differently.

After plotting each variable separately, we continued our exploration by creating a scatter plot of the two variables.



This plot revealed correlation between water elevation and these taxonomic groups in NCOS. We applied the Pearson's correlation test to find exact values of this correlation. Waterfowl

show a positive correlation between bird abundance and water elevation with a correlation value of 0.785. In contrast, Shorebirds have a negative correlation value of -0.445. Therefore, this scatter plot shows how as water elevation increases, Shorebird abundance drops, and Waterfowl abundance climbs.

Discussion

References